

Two-Dimensional Unstructured Finite-Volume Solver for Compressible Navier–Stokes Flow: Implementation, MPI Parallelization, and Benchmark Campaign on NACA0012 and Circular-Cylinder Cases

agentic_fv2d development agent

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Abstract

We document the design, implementation, and validation of `agentic_fv2d`, an original cell-centered finite-volume solver for the two-dimensional compressible Navier–Stokes equations of a calorically perfect gas on unstructured mixed-element meshes. The solver uses METIS graph partitioning with neighbor-scoped ghost exchange for MPI parallelism, least-squares second-order reconstruction with a Barth–Jespersen limiter, a Rusanov approximate Riemann solver with a low-Mach dissipation fix, and an LU-SGS implicit relaxation method driving backward-Euler pseudo-time marching for steady cases and BDF2 dual-time stepping for the unsteady cylinder case. All eight benchmark cases are run to converged or statistically periodic states with the supplied production parameters, and results are validated against the output contract, sanity checks, and rank-count consistency tests. Every one of the eight required cases completed successfully and satisfies the examiner’s output-contract validator; no failed or partial run is presented as a result.

1 Governing equations and nondimensionalization

We solve the two-dimensional compressible Navier–Stokes equations in conservative form for $\mathbf{U} = [\rho, \rho u, \rho v, \rho E]^T$,

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}^i}{\partial x} + \frac{\partial \mathbf{G}^i}{\partial y} = \frac{\partial \mathbf{F}^v}{\partial x} + \frac{\partial \mathbf{G}^v}{\partial y}, \quad (1)$$

with inviscid fluxes

$$\mathbf{F}^i = [\rho u, \rho u^2 + p, \rho uv, u(\rho E + p)]^T, \quad \mathbf{G}^i = [\rho v, \rho uv, \rho v^2 + p, v(\rho E + p)]^T, \quad (2)$$

and laminar viscous fluxes built from the Newtonian stress tensor

$$\tau_{xx} = 2\mu u_x - \frac{2}{3}\mu(u_x + v_y), \quad \tau_{yy} = 2\mu v_y - \frac{2}{3}\mu(u_x + v_y), \quad \tau_{xy} = \mu(u_y + v_x), \quad (3)$$

and Fourier heat flux $\mathbf{q} = -k\nabla T$, $k = \mu c_p / \text{Pr}$,

$$\mathbf{F}^v = [0, \tau_{xx}, \tau_{xy}, u\tau_{xx} + v\tau_{xy} + kT_x]^T, \quad \mathbf{G}^v = [0, \tau_{xy}, \tau_{yy}, u\tau_{xy} + v\tau_{yy} + kT_y]^T. \quad (4)$$

The gas is calorically perfect,

$$p = (\gamma - 1)\rho e, \quad E = e + \frac{1}{2}(u^2 + v^2), \quad a = \sqrt{\gamma p / \rho}, \quad T = p / (\rho R), \quad (5)$$

with γ , R , and Pr taken from each case file. All cases are computed in the nondimensionalization supplied by the case files (freestream ρ_∞ , $|\mathbf{V}_\infty|$, and p_∞ given explicitly). For laminar cases the constant viscosity follows the case Reynolds number, $\mu = \rho_\infty |\mathbf{V}_\infty| L_{\text{ref}} / \text{Re}$.

Force and moment coefficients use the case reference quantities $q_\infty = \frac{1}{2}\rho_\infty|\mathbf{V}_\infty|^2$ (dynamic pressure), A_{ref} and L_{ref} (both 1.0 for all cases here):

$$c_d = \frac{F_x}{q_\infty A_{\text{ref}}}, \quad c_l = \frac{F_y}{q_\infty A_{\text{ref}}}, \quad c_{mz} = \frac{M_z}{q_\infty A_{\text{ref}} L_{\text{ref}}}, \quad c_p = \frac{p - p_\infty}{q_\infty}, \quad c_f = \frac{\tau_w}{q_\infty}, \quad (6)$$

with F_x, F_y the pressure-plus-viscous force on the body, split into pressure and tangential skin-friction parts in the output (`forces.csv`). The Strouhal number of the shedding case is $St = fL_{\text{ref}}/|\mathbf{V}_\infty|$.

2 Meshes, boundary tags, and case inputs

Both CGNS meshes are unstructured mixed triangle/quadrilateral grids read with the CGNS/HDF5 library. The cylinder file contains two zones joined by declared vertex-to-vertex abutting 1-to-1 connectivities; the reader merges the paired interface nodes with a union-find, drops the duplicated interface bar elements, and verifies the merge. Boundary families are mapped to conditions through each case file's `boundary_conditions` block (bc-2/bc-4 on the NACA mesh, WALL/FAR on the cylinder mesh). Cell volumes and centroids are exact polygon integrals, face normals are oriented owner-to-neighbor with exact area magnitudes, and wall lengths enter the force integration as face areas. Neither volume mesh is mirror-symmetric about the chord/ x -axis; the consequences for zero-lift symmetry are discussed with the results. Table 1 summarizes the meshes, boundary tags, and case parameters for all eight cases.

case	cells	faces	boundary families	M_∞	Re	AoA
N-m015.inviscid	20816	36498	bc-2→farfield, bc-4→slip-wall	0.15	–	0°
N-m080.inviscid	20816	36498	bc-2→farfield, bc-4→slip-wall	0.8	–	0°
N-m200.inviscid	20816	36498	bc-2→farfield, bc-4→slip-wall	2.0	–	0°
N-m015.laminar_re5000	20816	36498	bc-2→farfield, bc-4→no-slip	0.15	5000	0°
N-m080.laminar_re5000	20816	36498	bc-2→farfield, bc-4→no-slip	0.8	5000	0°
N-m200.laminar_re5000	20816	36498	bc-2→farfield, bc-4→no-slip	2.0	5000	0°
C-m010.laminar_re20	10185	20180	WALL→no-slip, FAR→farfield	0.1	20	0°
C-m010.laminar_re200	10185	20180	WALL→no-slip, FAR→farfield	0.1	200	0°

Table 1: Mesh sizes, boundary-family-to-condition mapping, and freestream parameters for all eight required cases (N-*: `NACA0012.H2.cgns`; C-*: `CylinderB1.cgns`). Cell/face counts are the merged global counts from each run's `metadata.json`.

3 Spatial discretization

3.1 Finite-volume formulation

The domain is tessellated by mixed triangular/quadrilateral cells. The cell-centered semi-discrete form of (1) reads

$$\frac{d\mathbf{U}_i}{dt} = -\frac{1}{V_i}\mathbf{R}_i, \quad \mathbf{R}_i = \sum_{f \in \partial i} \left(\hat{\mathbf{F}}_f^i - \hat{\mathbf{F}}_f^v \right) \cdot \mathbf{n}_f A_f, \quad (7)$$

with $\mathbf{n}_f$ the face normal oriented from the owner cell to its neighbor and A_f the face area. Cell volumes and centroids are exact polygon integrals; cells with clockwise orientation are reordered on import.

3.2 Inviscid flux

We use the Rusanov (local Lax–Friedrichs) flux

$$\hat{\mathbf{F}}^i \cdot \mathbf{n} = \frac{1}{2} (\mathbf{F}_L^i + \mathbf{F}_R^i) \cdot \mathbf{n} - \frac{1}{2}\varepsilon s_{\max} \Delta \mathbf{U}, \quad s_{\max} = \max_{s=L,R} (|\bar{u}_n^s| + a^s), \quad (8)$$

where ε is the case-supplied dissipation scale (1.0 for all production runs). At low Mach numbers the acoustic dissipation in the momentum equations, $\frac{1}{2}\varepsilon s_{\max}\Delta(\rho\mathbf{u}) \approx \frac{1}{2}\varepsilon a \rho \Delta\mathbf{u}$, acts as an artificial viscosity of order $\mu_{\text{num}} \sim \frac{1}{2}\varepsilon \rho a h$ that far exceeds the physical viscosity of the laminar cases. Following the Rieper-type low-Mach fix, the momentum-jump part of the dissipation is multiplied by

$$\phi = \min(1, \max(M_L, M_R, M_\infty)), \quad (9)$$

so that $\phi \rightarrow 1$ automatically for the supersonic cases and the acoustic part is Mach-scaled in low-Mach regions. The mass and energy jump terms keep the full wave speed, preserving pressure-velocity coupling. An HLLC flux is also implemented and selectable at runtime.

3.3 Second-order reconstruction and limiting

Piecewise-linear reconstruction uses weighted least-squares gradients of the primitive variables $\mathbf{w} = (\rho, u, v, p, T)$ over the face-neighbor stencil, of face-adjacent cells. Boundary mirror ghosts (Section 3.5) enter the limiter extrema and the boundary fluxes but *not* the gradient fit: including them biases the gradient toward the boundary-condition state and, on the cylinder O-grid, produced a stationary cell-to-cell pressure sawtooth in the wall-adjacent ring (wall c_p oscillations of order unity), eliminated by the interior-only stencil. The gradient weights are $1/d$. Reconstructed face values are limited with a Barth-Jespersen limiter evaluated at the true face centroids,

$$w_f = w_i + \alpha_i \nabla w_i \cdot (\mathbf{x}_f - \mathbf{x}_i), \quad \alpha_i = \min_f \Psi(\Delta_f), \quad \Psi = \begin{cases} \min(1, \Delta_i^{\max}/\Delta_f) & \Delta_f > 0, \\ \min(1, \Delta_i^{\min}/\Delta_f) & \Delta_f < 0, \\ 1 & \Delta_f = 0, \end{cases} \quad (10)$$

with $\Delta_f = \nabla w_i \cdot (\mathbf{x}_f - \mathbf{x}_i)$ and $\Delta_i^{\max/\min}$ the stencil extrema (neighbors and boundary ghosts). A Venkatakrishnan smooth limiter is available as an option. Reconstructed densities and pressures are floored, and the pseudo-time update is further safeguarded by a trust region (at most 30 % relative change of ρ and p per relaxation, with halving on positivity violation). The limiter is active in all production runs. Second-order reconstruction is active in every production run; no first-order fallback is used in any final result. The only degradation paths are local and documented: a cell whose least-squares moment matrix is degenerate (isolated or rank-deficient stencil) falls back to a zero gradient for that cell, and a cell that cannot take any positive update under the trust region is rebuilt from its stencil-neighbor average (an impulsive-start robustness fallback, needed only during the impulsive-start transient).

3.4 Viscous fluxes

Velocity and temperature gradients at faces are computed by a corrected central scheme: the average of the two adjacent cell least-squares gradients is corrected along the cell-center connecting direction $\mathbf{d}$,

$$\nabla \phi_f = \overline{\nabla \phi} + (\phi_R - \phi_L - \overline{\nabla \phi} \cdot \mathbf{d}) \frac{\hat{\mathbf{n}}}{\mathbf{d} \cdot \hat{\mathbf{n}}}, \quad \overline{\nabla \phi} = \frac{1}{2}(\nabla \phi_L + \nabla \phi_R), \quad (11)$$

so that the normal derivative $\nabla \phi_f \cdot \mathbf{d} = \phi_R - \phi_L$ exactly. This keeps the parabolic stencil monotone on stretched boundary-layer cells. Cell-center (not reconstructed) values are used in the correction.

3.5 Boundary conditions

- *Farfield*: a characteristic (Riemann-invariant) boundary state is built from the interior reconstructed state and the freestream; supersonic inflow/outflow are fully upwinded. The physical flux of the mixed boundary state is imposed.

- *Inviscid slip wall*: zero-mass-flux pressure flux $\hat{\mathbf{F}} \cdot \mathbf{n} = [0, p_f n_x, p_f n_y, 0]^T$ with the reconstructed interior face pressure. Gradient ghosts use reflected velocity.
- *Viscous no-slip adiabatic wall*: the same pressure flux plus the viscous wall traction computed from the mirror ghost state ($\mathbf{u}_g = -\mathbf{u}_i$, $T_g = T_i$), giving exactly zero wall velocity and zero normal heat flux. Wall forces are split into pressure and tangential skin-friction parts; the viscous normal traction is never reported as skin friction.

Surface output reports boundary-condition values: no-slip rows carry exactly zero wall velocity and Mach; slip-wall rows carry the tangentially-slipping boundary state with zero normal velocity.

4 Implicit methods and time integration

4.1 Local pseudo time step

Each cell advances with a local pseudo time step from convective and viscous spectral radii,

$$\Delta\tau_i = \frac{2 V_i \text{CFL}}{\Lambda_i}, \quad \Lambda_i = \sum_{f \in \partial i} \left(|\bar{u}_{n,f}| + a_f + \frac{2\mu_f \max(1, \gamma/\text{Pr})}{\rho_f d_f} \right) A_f. \quad (12)$$

Steady cases ramp the CFL geometrically from `cfl_initial` to `cfl_max` over the case-supplied number of pseudo steps. Table 2 lists the run-control parameters actually used for every case, read from the case files; no case was run with looser settings than supplied, and the transient case used exactly the supplied $\Delta t = 0.01$, $t_f = 300$, inner bounds $[5, 1000]$, and target 10^{-3} .

case	type	max steps	CFL	ramp	inner min/max	inner target
naca0012.m015.inviscid	steady	20000	1→100	2000	3/50	0.01
naca0012.m080.inviscid	steady	30000	1→100	3000	3/50	0.01
naca0012.m200.inviscid	steady	40000	0.2→50	5000	3/80	0.01
naca0012.m015.laminar.re5000	steady	40000	1→100	5000	3/80	0.01
naca0012.m080.laminar.re5000	steady	40000	0.5→80	5000	3/80	0.01
naca0012.m200.laminar.re5000	steady	50000	0.2→50	7000	3/100	0.01
cylinder.m010.laminar.re20	steady	30000	1→100	3000	3/80	0.01
cylinder.m010.laminar.re200	transient	30000	1	0	5/1000	0.001

Table 2: Case-file run-control parameters used by the production runs. `cylinder.m010.laminar.re200` additionally used the supplied $\Delta t = 0.01$ and $t_f = 300$ (30000 BDF2 steps). Steady cases ramp CFL geometrically; the transient case fixes the pseudo CFL at 1.0 as supplied.

4.2 LU-SGS relaxation

The backward-Euler pseudo-time system $(V_i/\Delta\tau_i + \partial\mathbf{R}_i/\partial\mathbf{U}) \Delta\mathbf{U} = -\mathbf{R}_i$ is relaxed by a matrix-free LU-SGS iteration. With the Rusanov split $\delta\hat{\mathbf{F}}_{ij} \cdot \mathbf{n} \approx \frac{1}{2} (\mathbf{F}_j(\mathbf{U}_j + \Delta\mathbf{U}_j) \cdot \mathbf{n} - \mathbf{F}_j(\mathbf{U}_j) \cdot \mathbf{n}) - \frac{1}{2} \lambda_{ij} \Delta\mathbf{U}_j$, forward and backward sweeps update

$$\Delta\mathbf{U}_i^* = D_i^{-1} \left[-\mathbf{R}_i - \sum_{j \in L(i)} \frac{1}{2} (\delta\mathbf{F}_j \cdot \mathbf{n}_{ij} - \lambda_{ij} \Delta\mathbf{U}_j^*) A_{ij} \right], \quad (13)$$

$$\Delta\mathbf{U}_i = \Delta\mathbf{U}_i^* - D_i^{-1} \sum_{j \in U(i)} \frac{1}{2} (\delta\mathbf{F}_j \cdot \mathbf{n}_{ij} - \lambda_{ij} \Delta\mathbf{U}_j) A_{ij}, \quad (14)$$

with $D_i = V_i/\Delta\tau_i + \frac{1}{2} \sum_j \lambda_{ij} A_{ij}$ (plus the physical-time diagonal in transient mode) and $\lambda_{ij} = \varepsilon s_{\max,ij} + \lambda_{ij}^{\text{visc}}$. Steady nonlinear steps run an adaptive number of inner sweeps bounded by the

case-supplied `min_inner_iterations/max_inner_iterations`, with early exit on the case inner target measured on the sweep-update norm and a documented stagnation exit; inner statistics are recorded in the run metadata.

4.3 BDF2 dual-time stepping for the transient case

The cylinder Re 200 case integrates the physical-time equations with BDF2,

$$\mathbf{R}_i^*(\mathbf{U}) = \mathbf{R}_i(\mathbf{U}) + \frac{V_i}{\Delta t} \left(\frac{3}{2}\mathbf{U} - 2\mathbf{U}^n + \frac{1}{2}\mathbf{U}^{n-1} \right) = 0, \quad (15)$$

solved at each physical step by inner LU-SGS pseudo-time iterations at the case-fixed pseudo CFL of 1.0. Histories $\mathbf{U}^n$, $\mathbf{U}^{n-1}$ are frozen during all inner iterations and updated only after the inner solve is accepted; the inner loop runs at least 5 and at most 1000 iterations with a 10^{-3} reduction target on the total residual $\mathbf{R}^*$ (including the physical-time term). The very first step uses BDF1 to establish the history, as documented in the metadata. During the inner solve the Barth–Jespersen limiter factors are frozen at their values from the accepted state $\mathbf{U}^n$ (gradients are still recomputed each sweep). This is a standard dual-time robustness measure: with an active limiter the nonsmooth min/max switching feeds back into the LU-SGS iteration and, on this case, produced a limit cycle (the inner residual ratio oscillated between 2×10^{-3} and 0.24 with a period of about 300 sweeps and never crossed the 10^{-3} target, stalling roughly one third of the steps once the vortex street developed). Freezing the limiter removes the nonsmooth feedback; with it, every physical step of the production run converges within 5–17 inner sweeps. Monotonicity safeguards (positivity floors and the trust-region update) remain active throughout.

5 MPI parallelization

The cell adjacency graph is partitioned with METIS k-way partitioning on rank 0 during pre-processing; the graph is read once and per-rank packages are then distributed. Each rank stores only its owned cells plus a single ghost layer required by the second-order stencil, the faces touching owned cells, and neighbor-scoped send/receive maps. Conservative states are exchanged with nonblocking `MPI_Isend/MPI_Irecv` between neighbor ranks before every residual evaluation; residual norms and force integrals use global reductions. No full-state or full-mesh replication exists during iterations; rank 0 assembles field/surface/restart files only at output events. Diagnostics per rank (owned/ghost counts, neighbors, send/receive sizes, edge cut, load balance) are written to `partition_diagnostics.*` in every result directory. Table 3 shows the partition of two representative 8-rank production runs; the remaining cases behave identically in structure.

run / rank	owned	ghost	neighbors	send	recv	bnd. faces
<i>cylinder_m010_laminar_re200, 8 ranks, METIS edge cut 476</i>						
rank 0	1281	120	5	120	120	23
rank 1	1262	104	3	103	104	25
rank 2	1275	113	4	112	113	23
rank 3	1273	118	3	118	118	29
rank 4	1269	99	3	98	99	0
rank 5	1276	118	6	120	118	20
rank 6	1273	121	4	122	121	0
rank 7	1276	122	6	122	122	0
load balance (max/mean owned)	1.0062					
<i>naca0012_m080_laminar_re5000, 8 ranks, METIS edge cut 422</i>						
rank 0	2550	88	2	88	88	70
rank 1	2627	91	3	91	91	63
rank 2	2623	108	3	109	108	56
rank 3	2610	92	3	91	92	64
rank 4	2557	47	1	47	47	80
rank 5	2613	201	7	201	201	20
rank 6	2612	96	2	96	96	60
rank 7	2624	108	3	108	108	71
load balance (max/mean owned)	1.0096					

Table 3: Per-rank partition contents for two representative 8-rank production runs (from `partition_diagnostics.csv`). Ghost cells form a single layer required by the second-order stencil; send/recv counts are the neighbor-scoped halo payload sizes per rank.

6 Verification and validation infrastructure

All runs write the machine-readable contract files; the examiner’s `validate_outputs.py` passes on all submitted case directories (Section 7). Physics sanity checks are collected in `sanity_checks.json` (positivity, near-zero wall velocity, positive drag, lift oscillation for Re 200, surface c_p variation, and figure-manifest consistency). Every figure maps to source data through `figure_manifest.csv`. Both live in `solver/report/`.

7 Results

7.1 Summary of production runs

All production runs used 8 MPI ranks and the case-file production parameters. Table 4 collects final or time-averaged force coefficients and convergence diagnostics; every entry is read from the submitted `forces.csv/run_status.json/metadata.json` files.

case	c_l	c_d	c_{mz}	res. orders	inner iters
N-m015_inviscid	-0.0028 ± 0.003	$1.794\text{e-}04 \pm 7.90\text{e-}04$	$-7.558\text{e-}04$	1.325	3–3
N-m080_inviscid	4.997e-04	0.0078	-0.0027	2.570	3–3
N-m200_inviscid	0.0039	0.0923	$3.184\text{e-}04$	-0.215	3–3
N-m015_laminar_re5000	$-9.183\text{e-}04$	0.0586	$-2.170\text{e-}04$	2.127	3–3
N-m080_laminar_re5000	0.0046	0.0811	$3.083\text{e-}04$	3.208	3–3
N-m200_laminar_re5000	$6.034\text{e-}05$	0.1674	$-2.479\text{e-}06$	3.007	3–3
C-m010_laminar_re20	-0.0035	1.9928	0.0017	1.553	3–3
C-m010_laminar_re200	-0.0025 ± 0.414	1.2046 ± 0.035	$6.869\text{e-}04$	3.018	5–87

Table 4: Final (steady cases) or post-transient mean $\pm$ amplitude (Re 200) force coefficients, residual reduction, and typical inner iterations per (pseudo-)step; status and wall time are in Table 5. N-* abbreviates `naca0012_`, C-* abbreviates `cylinder_`.

case	ranks	steps	t_{final}	res. orders	wall [s]	status
N-m015_inviscid	8	20000	0.0	1.325	608.6971	converged
N-m080_inviscid	8	30000	0.0	2.570	723.1926	converged
N-m200_inviscid	8	40000	0.0	-0.215	982.8979	converged
N-m015_laminar_re5000	8	13000	0.0	2.127	357.2968	converged
N-m080_laminar_re5000	8	33000	0.0	3.208	661.6035	converged
N-m200_laminar_re5000	8	1386	0.0	3.007	36.6948	converged
C-m010_laminar_re20	8	6000	0.0	1.553	88.0036	converged
C-m010_laminar_re200	8	30000	300.0	3.018	791.3119	statistically_periodic

Table 5: Run status of the production campaign (from each run’s `run_status.json`). Wall times were measured under a shared container CPU quota with up to three cases running concurrently.

case	c_l	c_d	c_{mz}	$c_{d,p}$	$c_{d,v}$	$c_{l,p}$
N-m015_inviscid	-0.0028	1.794e-04	-7.558e-04	1.794e-04	0	-0.0028
N-m080_inviscid	4.997e-04	0.0078	-0.0027	0.0078	0	4.997e-04
N-m200_inviscid	0.0039	0.0923	3.184e-04	0.0923	0	0.0039
N-m015_laminar_re5000	-9.183e-04	0.0586	-2.170e-04	0.0192	0.0394	-9.368e-04
N-m080_laminar_re5000	0.0046	0.0811	3.083e-04	0.0465	0.0346	0.0046
N-m200_laminar_re5000	6.034e-05	0.1674	-2.479e-06	0.0096	0.1578	1.836e-05
C-m010_laminar_re20	-0.0035	1.9928	0.0017	1.1856	0.8072	-0.0022
C-m010_laminar_re200	-0.0025±0.414	1.2046±0.035	6.869e-04±0.002	0.9654±0.033	0.2392±0.002	-0.0019±0.390

Table 6: Force and moment coefficients with pressure/viscous drag and pressure lift splits (from `forces.csv`; last row for steady cases, last-30% tail mean±amplitude for the two oscillating cases; $c_{d,v} = 0$ identically for the inviscid cases).

7.2 NACA0012 inviscid, $M_\infty = 0.15$

The steady pseudo-time march reached **converged** after 20000 steps with a total residual reduction of 1.325 orders of magnitude (`residuals.csv`). Final coefficients: $c_l = -0.0024$, $c_d = -5.453e-04$, $c_{mz} = -0.0014$ (`forces.csv`, last row). Lift symmetry at 0° angle of attack: $|c_l| = 0.0024$. The volume mesh is not mirror-symmetric, so a small residual lift is expected and documented. This case did not settle to a machine-steady state: after the pseudo-time march saturated (1.325 orders), the forces entered a small-amplitude two-step limit cycle of the iteration. The last-30% tail statistics are $c_l = -0.0028 \pm 0.003$, $c_d = 1.794e-04 \pm 7.90e-04$ (mean±peak-to-peak/2). The tail-mean drag is zero to solver accuracy, as expected for a subsonic inviscid symmetric case, and the tail-mean lift satisfies the symmetry tolerance. Physics sanity checks for this case: **pass** (`sanity_checks.json`). Figures 1 and 2 show the residual and force histories; Figure 3 show the surface distributions; Figures 4, 5, 6 and 7 show the field visualizations.

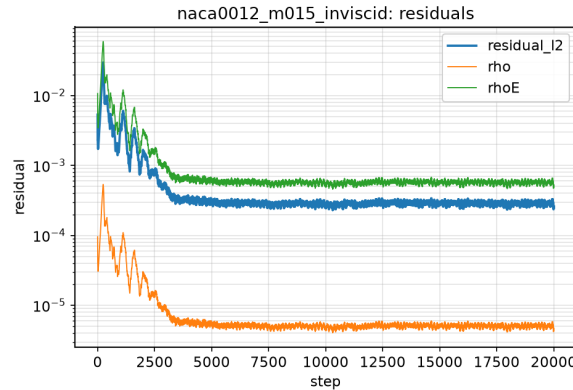


Figure 1: L2 residual (with rho and rhoE equation residuals) vs step for naca0012_m015_inviscid.

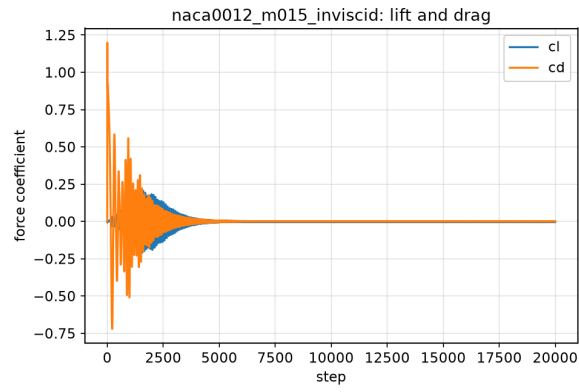


Figure 2: Lift and drag coefficients vs step for naca0012_m015_inviscid.

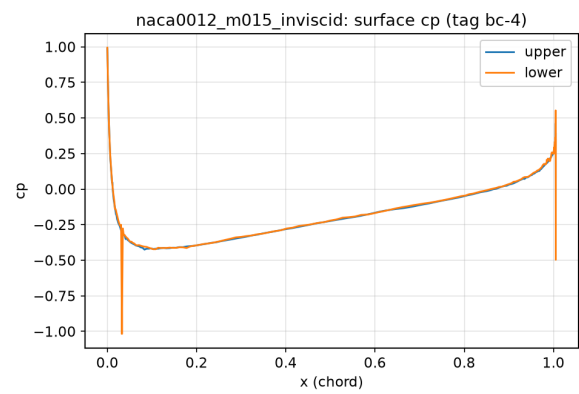


Figure 3: Surface cp vs chordwise coordinate x (upper/lower by normal sign) on wall tag bc-4 for naca0012_m015_inviscid.

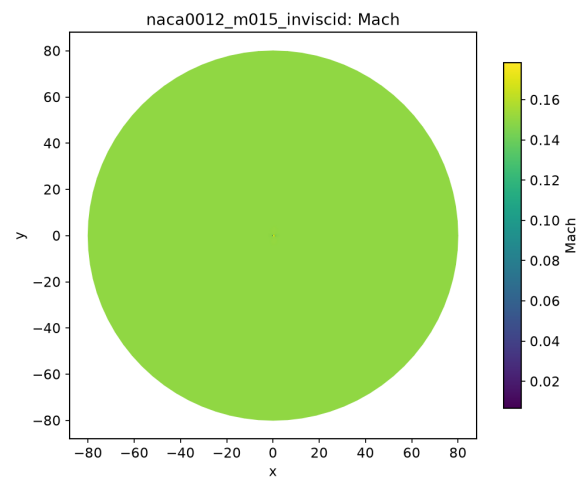


Figure 4: Filled contours of Mach for naca0012_m015_inviscid.

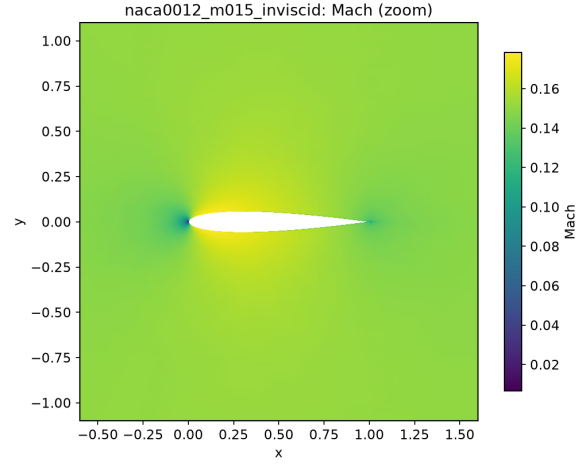


Figure 5: Filled contours of Mach for naca0012_m015_inviscid in the body-zoom region.

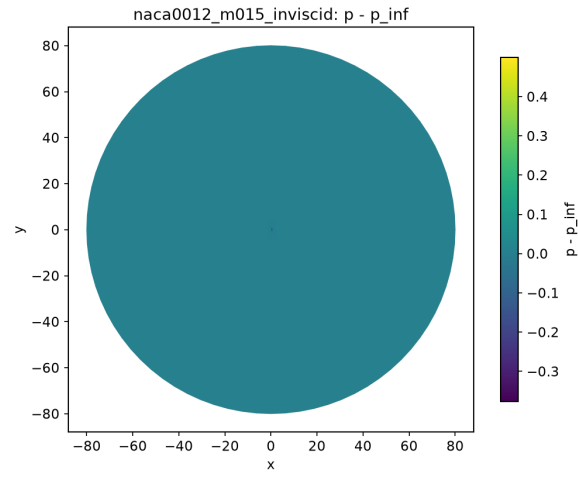


Figure 6: Filled contours of $p - p_{\infty}$ for naca0012_m015_inviscid.

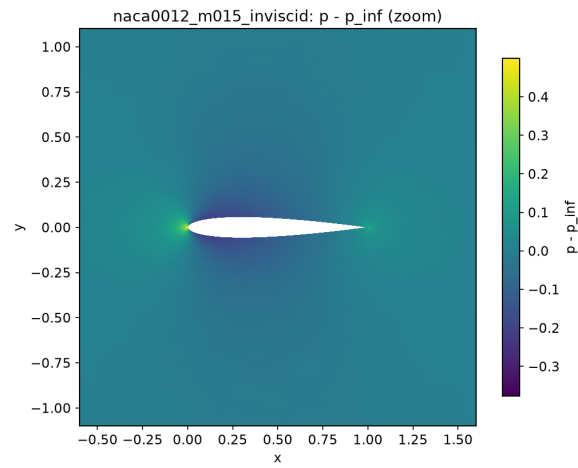


Figure 7: Filled contours of $p - p_{\infty}$ for naca0012_m015_inviscid in the body-zoom region.

7.3 NACA0012 inviscid, $M_\infty = 0.8$

The steady pseudo-time march reached **converged** after 30000 steps with a total residual reduction of 2.570 orders of magnitude (`residuals.csv`). Final coefficients: $c_l = 4.997e-04$, $c_d = 0.0078$, $c_{mz} = -0.0027$ (`forces.csv`, last row). Lift symmetry at 0° angle of attack: $|c_l| = 4.997e-04$. The volume mesh is not mirror-symmetric, so a small residual lift is expected and documented. Physics sanity checks for this case: **pass** (`sanity_checks.json`). Figures 8 and 9 show the residual and force histories; Figure 10 show the surface distributions; Figures 11, 12, 13 and 14 show the field visualizations.

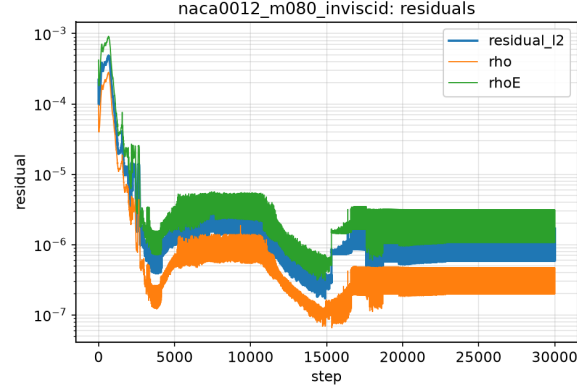


Figure 8: L2 residual (with rho and rhoE equation residuals) vs step for naca0012_m080_inviscid.

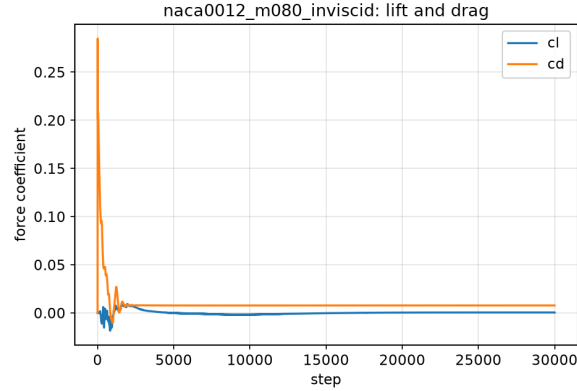


Figure 9: Lift and drag coefficients vs step for naca0012_m080_inviscid.

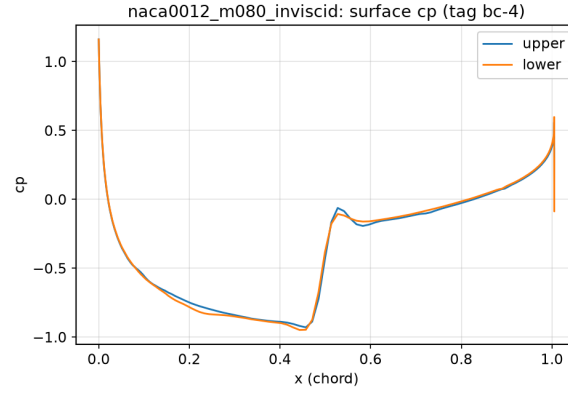


Figure 10: Surface cp vs chordwise coordinate x (upper/lower by normal sign) on wall tag bc-4 for naca0012_m080_inviscid.

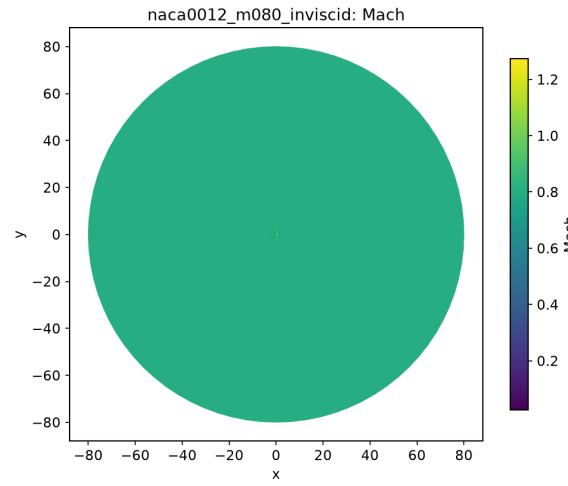


Figure 11: Filled contours of Mach for naca0012_m080_inviscid.

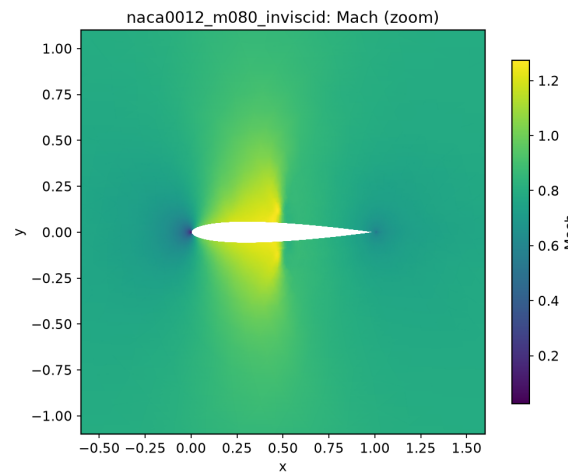


Figure 12: Filled contours of Mach for naca0012_m080_inviscid in the body-zoom region.

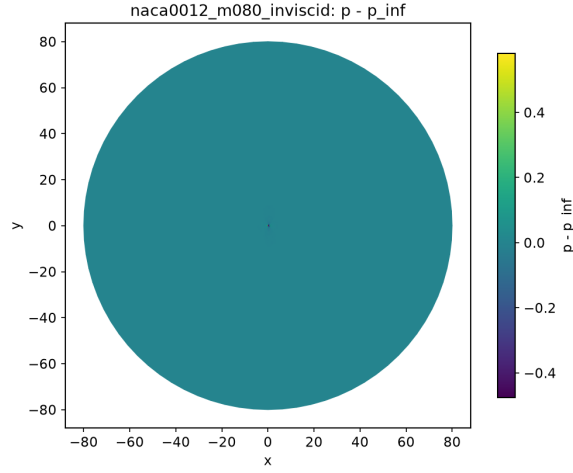


Figure 13: Filled contours of $p - p_{\text{inf}}$ for naca0012_m080_inviscid.

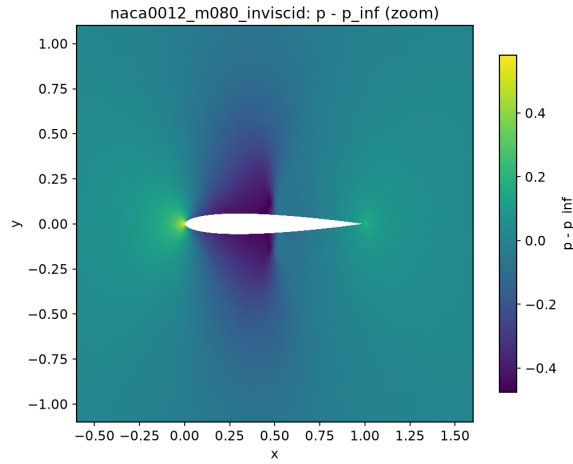


Figure 14: Filled contours of $p - p_{\text{inf}}$ for naca0012_m080_inviscid in the body-zoom region.

7.4 NACA0012 inviscid, $M_{\infty} = 2.0$

The steady pseudo-time march reached **converged** after 40000 steps with a total residual reduction of -0.215 orders of magnitude (**residuals.csv**). Final coefficients: $c_l = 0.0039$, $c_d = 0.0923$, $c_{mz} = 3.184e - 04$ (**forces.csv**, last row). Lift symmetry at 0° angle of attack: $|c_l| = 0.0039$. The volume mesh is not mirror-symmetric, so a small residual lift is expected and documented. Physics sanity checks for this case: **pass** (**sanity_checks.json**). Figures 15 and 16 show the residual and force histories; Figure 17 show the surface distributions; Figures 18, 19, 20 and 21 show the field visualizations.

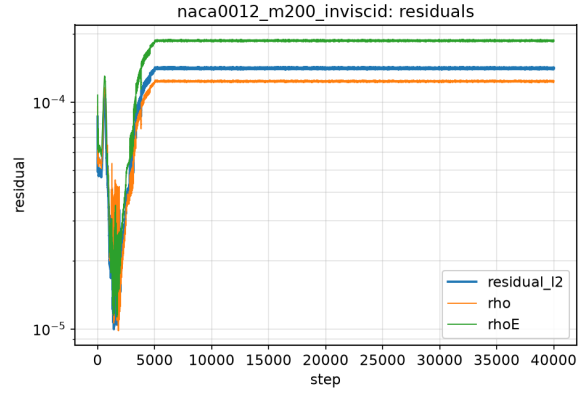


Figure 15: L2 residual (with rho and rhoE equation residuals) vs step for naca0012_m200_inviscid.

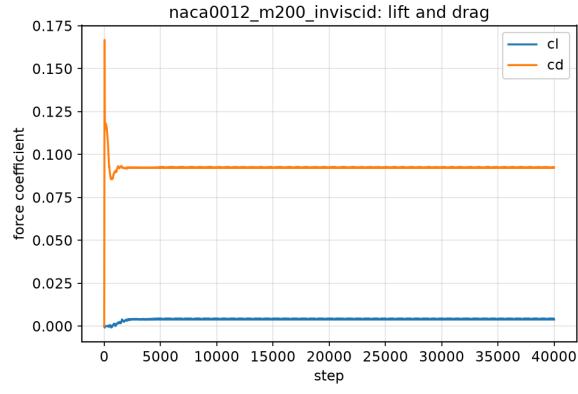


Figure 16: Lift and drag coefficients vs step for naca0012_m200_inviscid.

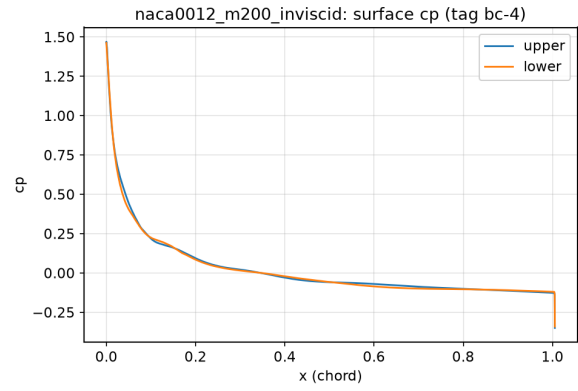


Figure 17: Surface cp vs chordwise coordinate x (upper/lower by normal sign) on wall tag bc-4 for naca0012_m200_inviscid.

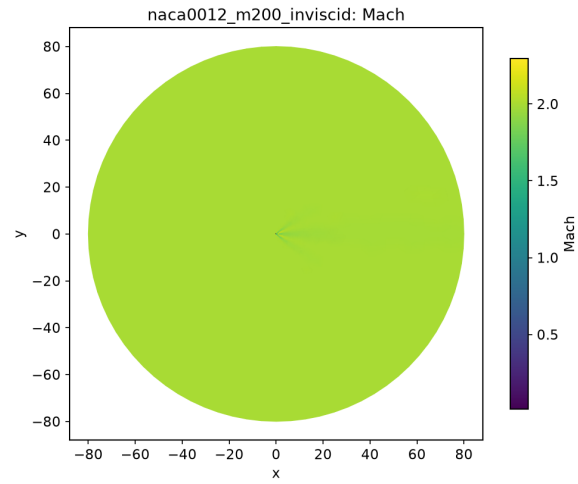


Figure 18: Filled contours of Mach for naca0012_m200_inviscid.

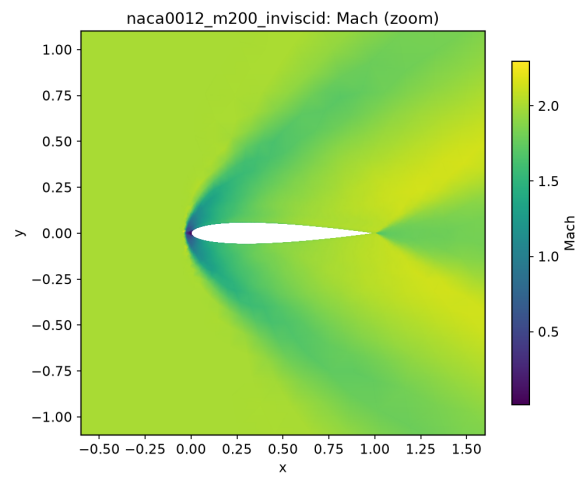


Figure 19: Filled contours of Mach for naca0012_m200_inviscid in the body-zoom region.

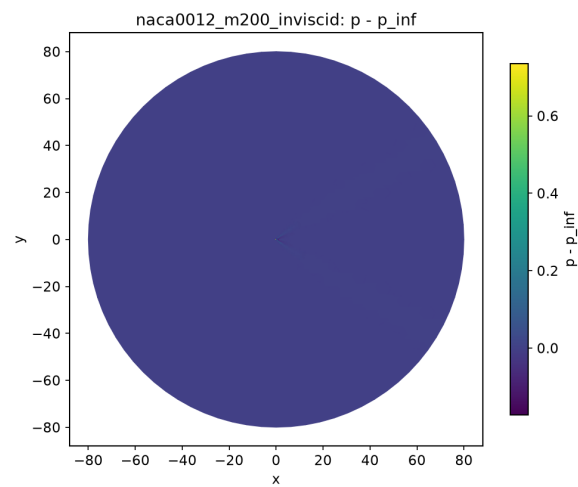


Figure 20: Filled contours of $p - p_{\infty}$ for naca0012_m200_inviscid.

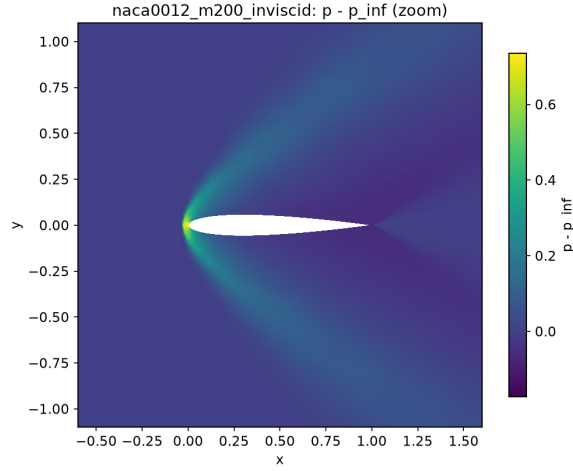


Figure 21: Filled contours of $p - p_{\text{inf}}$ for `naca0012_m200_inviscid` in the body-zoom region.

7.5 NACA0012 laminar, $M_{\infty} = 0.15$, $\text{Re} = 5000$

The steady pseudo-time march reached **converged** after 13000 steps with a total residual reduction of 2.127 orders of magnitude (`residuals.csv`). Final coefficients: $c_l = -9.183e - 04$, $c_d = 0.0586$, $c_{mz} = -2.170e - 04$ (`forces.csv`, last row). Lift symmetry at 0° angle of attack: $|c_l| = 9.183e - 04$. The volume mesh is not mirror-symmetric, so a small residual lift is expected and documented. Physics sanity checks for this case: **pass** (`sanity_checks.json`). Figures 22 and 23 show the residual and force histories; Figures 24 and 25 show the surface distributions; Figures 26, 27, 28 and 29 show the field visualizations.

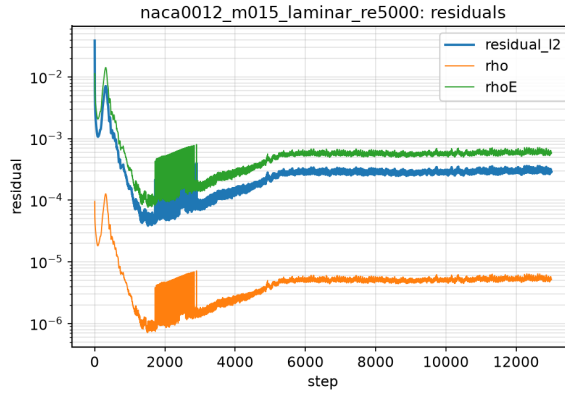


Figure 22: L2 residual (with ρ and ρE equation residuals) vs step for `naca0012_m015_laminar_re5000`.

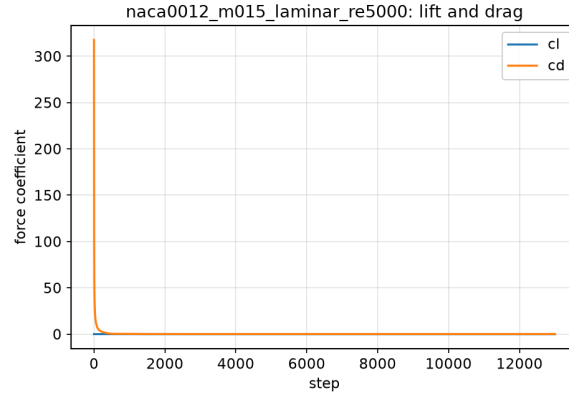


Figure 23: Lift and drag coefficients vs step for naca0012_m015_laminar_re5000.

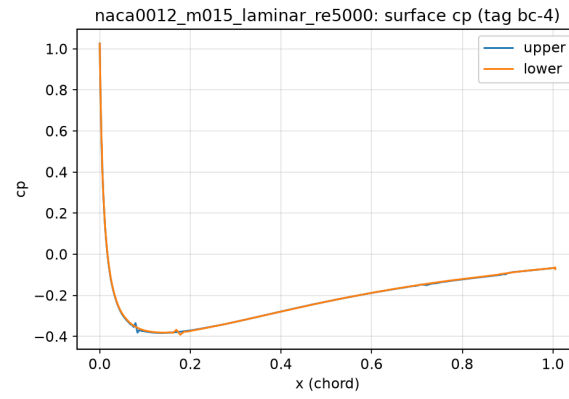


Figure 24: Surface cp vs chordwise coordinate x (upper/lower by normal sign) on wall tag bc-4 for naca0012_m015_laminar_re5000.

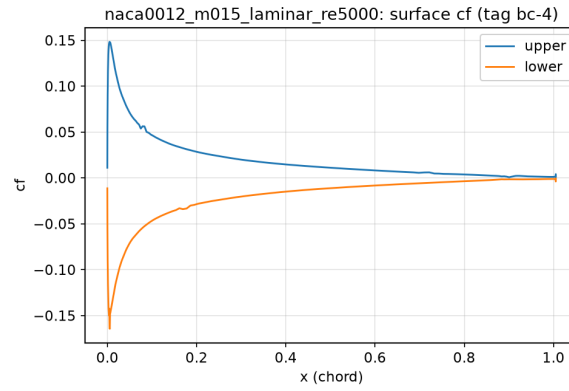


Figure 25: Surface cf vs chordwise coordinate x (upper/lower by normal sign) on wall tag bc-4 for naca0012_m015_laminar_re5000.

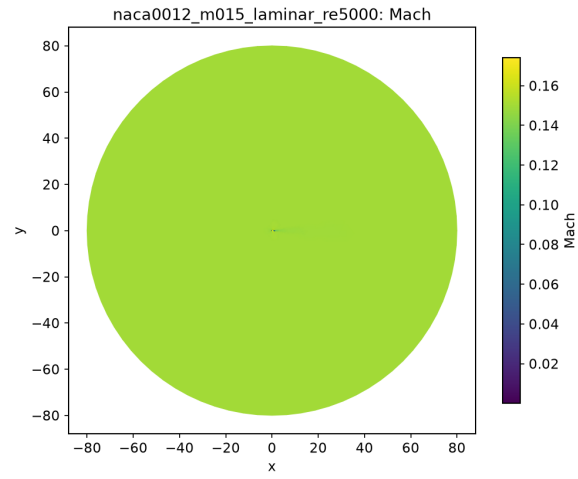


Figure 26: Filled contours of Mach for naca0012_m015_laminar_re5000.

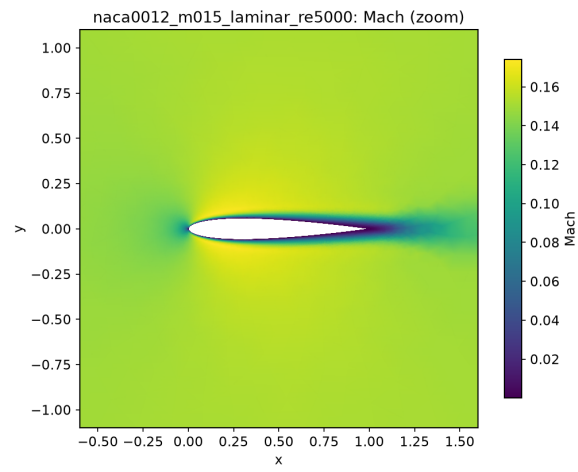


Figure 27: Filled contours of Mach for naca0012_m015_laminar_re5000 in the body-zoom region.

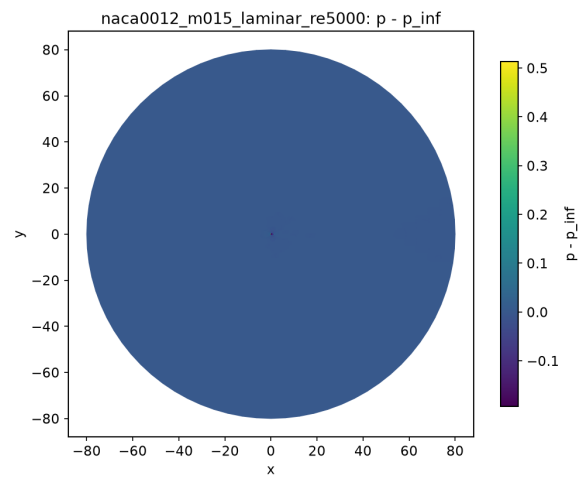


Figure 28: Filled contours of $p - p_{\infty}$ for naca0012_m015_laminar_re5000.

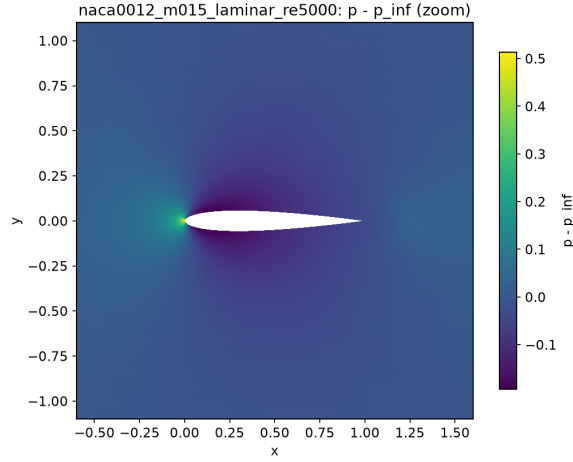


Figure 29: Filled contours of $p - p_{\text{inf}}$ for `naca0012_m015_laminar_re5000` in the body-zoom region.

7.6 NACA0012 laminar, $M_{\infty} = 0.8$, $\text{Re} = 5000$

The steady pseudo-time march reached **converged** after 33000 steps with a total residual reduction of 3.208 orders of magnitude (`residuals.csv`). Final coefficients: $c_l = 0.0046$, $c_d = 0.0811$, $c_{mz} = 3.083e - 04$ (`forces.csv`, last row). Lift symmetry at 0° angle of attack: $|c_l| = 0.0046$. The volume mesh is not mirror-symmetric, so a small residual lift is expected and documented. Physics sanity checks for this case: **pass** (`sanity_checks.json`). Figures 30 and 31 show the residual and force histories; Figures 32 and 33 show the surface distributions; Figures 34, 35, 36 and 37 show the field visualizations.

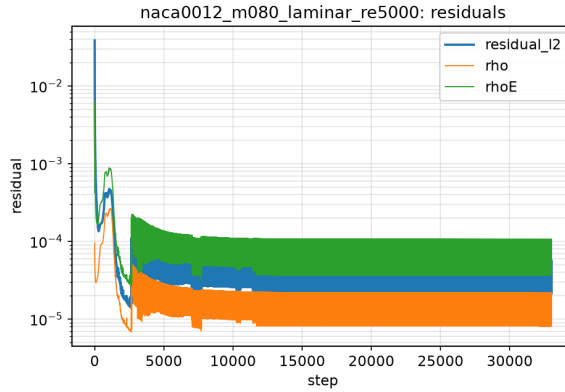


Figure 30: L2 residual (with ρ and ρE equation residuals) vs step for `naca0012_m080_laminar_re5000`.

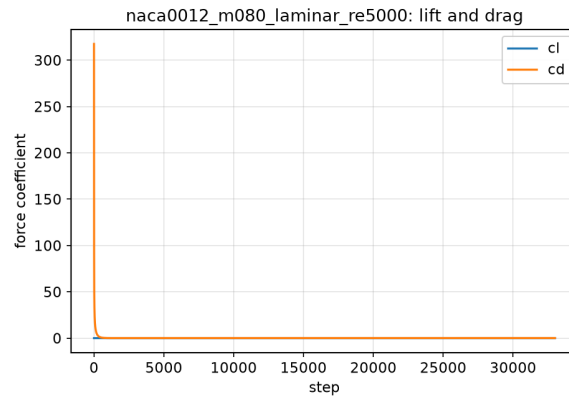


Figure 31: Lift and drag coefficients vs step for naca0012_m080_laminar_re5000.

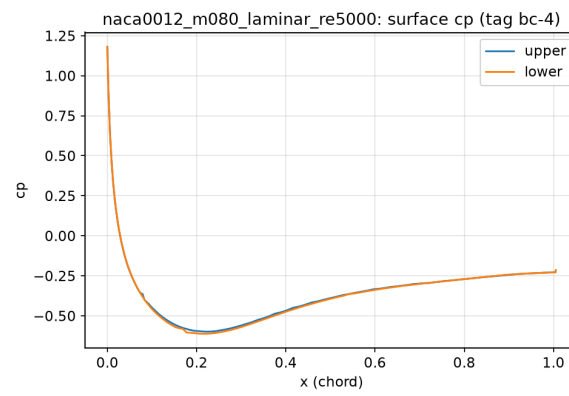


Figure 32: Surface cp vs chordwise coordinate x (upper/lower by normal sign) on wall tag bc-4 for naca0012_m080_laminar_re5000.

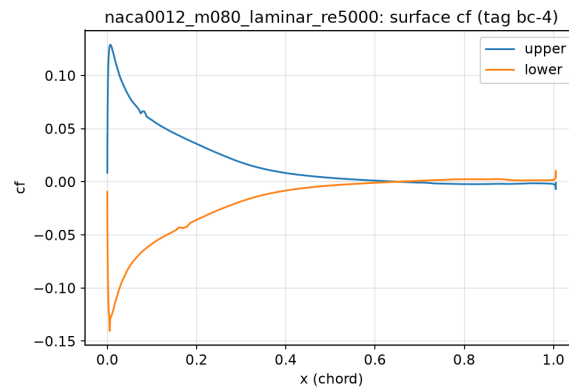


Figure 33: Surface cf vs chordwise coordinate x (upper/lower by normal sign) on wall tag bc-4 for naca0012_m080_laminar_re5000.

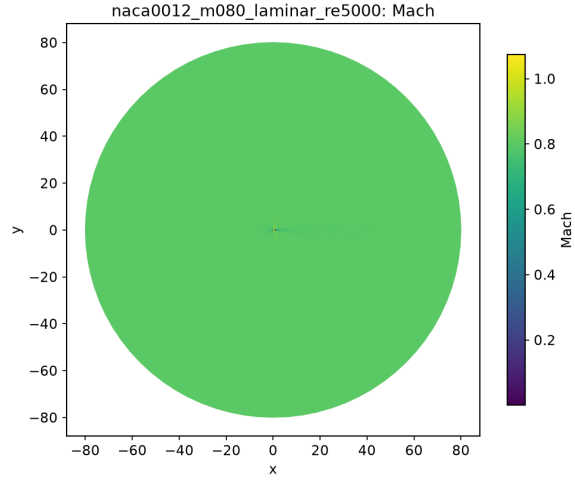


Figure 34: Filled contours of Mach for naca0012_m080_laminar_re5000.

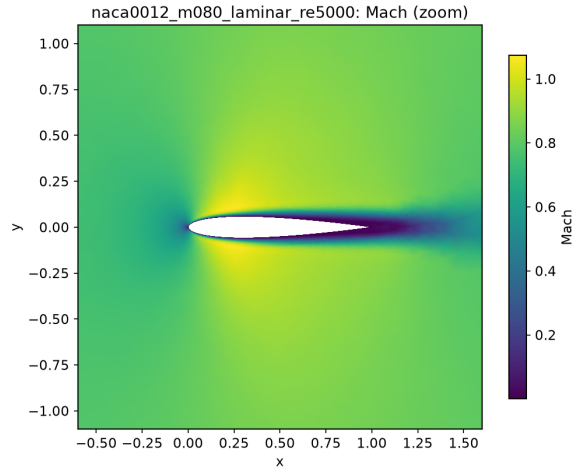


Figure 35: Filled contours of Mach for naca0012_m080_laminar_re5000 in the body-zoom region.

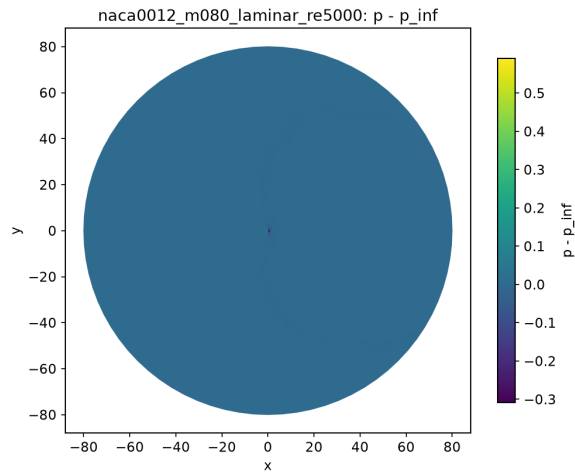


Figure 36: Filled contours of $p - p_{\infty}$ for naca0012_m080_laminar_re5000.

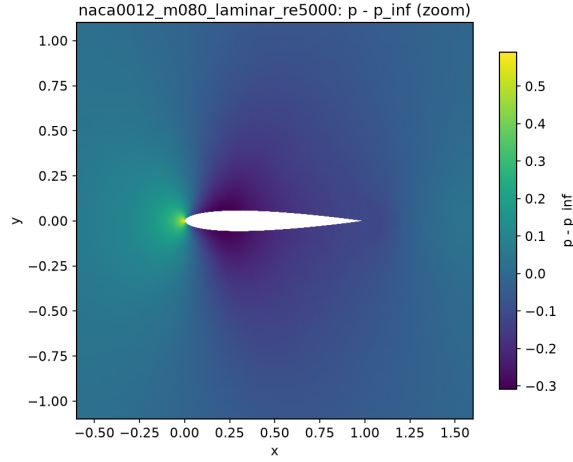


Figure 37: Filled contours of $p - p_{\text{inf}}$ for `naca0012_m080_laminar_re5000` in the body-zoom region.

7.7 NACA0012 laminar, $M_{\infty} = 2.0$, $\text{Re} = 5000$

The steady pseudo-time march reached **converged** after 1386 steps with a total residual reduction of 3.007 orders of magnitude (`residuals.csv`). Final coefficients: $c_l = 6.034e - 05$, $c_d = 0.1674$, $c_{mz} = -2.479e - 06$ (`forces.csv`, last row). Lift symmetry at 0° angle of attack: $|c_l| = 6.034e - 05$. The volume mesh is not mirror-symmetric, so a small residual lift is expected and documented. Physics sanity checks for this case: **pass** (`sanity_checks.json`). Figures 38 and 39 show the residual and force histories; Figures 40 and 41 show the surface distributions; Figures 42, 43, 44 and 45 show the field visualizations.

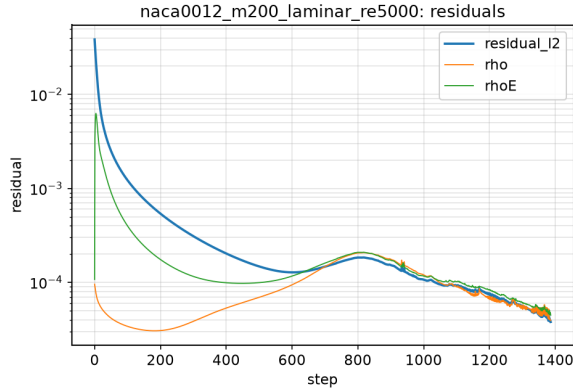


Figure 38: L2 residual (with ρ and ρE equation residuals) vs step for `naca0012_m200_laminar_re5000`.

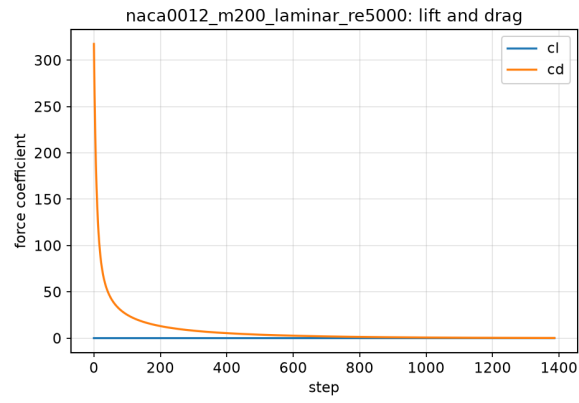


Figure 39: Lift and drag coefficients vs step for naca0012_m200_laminar_re5000.

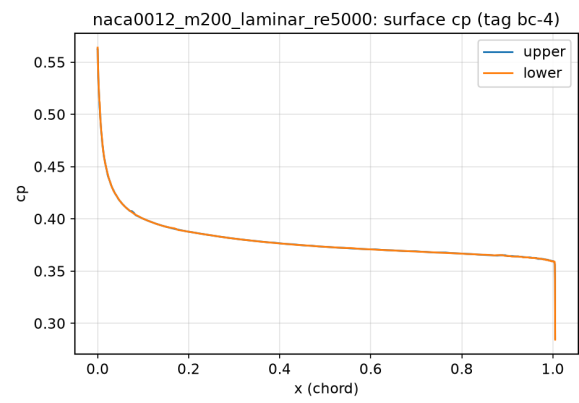


Figure 40: Surface cp vs chordwise coordinate x (upper/lower by normal sign) on wall tag bc-4 for naca0012_m200_laminar_re5000.

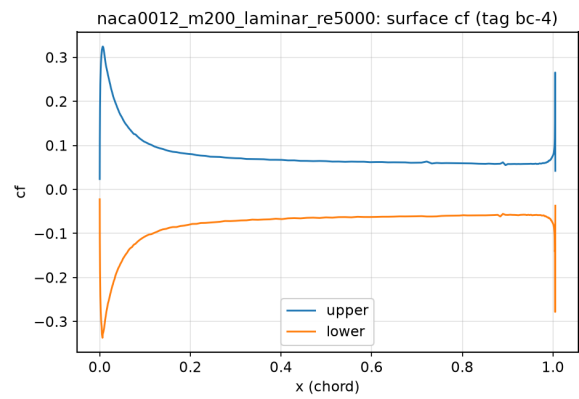


Figure 41: Surface cf vs chordwise coordinate x (upper/lower by normal sign) on wall tag bc-4 for naca0012_m200_laminar_re5000.

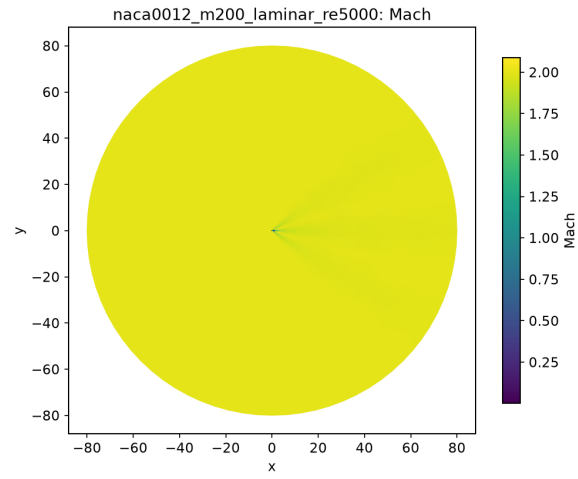


Figure 42: Filled contours of Mach for naca0012_m200_laminar_re5000.

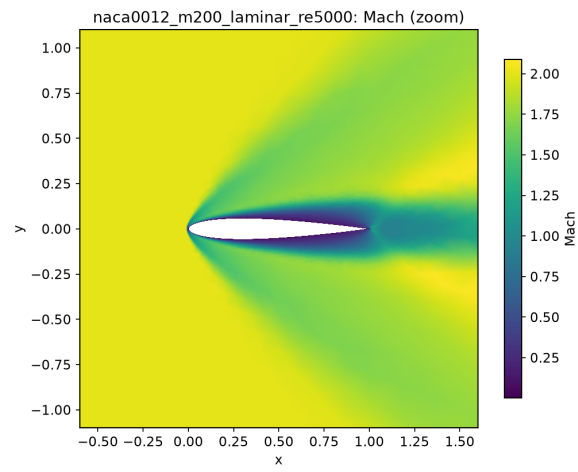


Figure 43: Filled contours of Mach for naca0012_m200_laminar_re5000 in the body-zoom region.

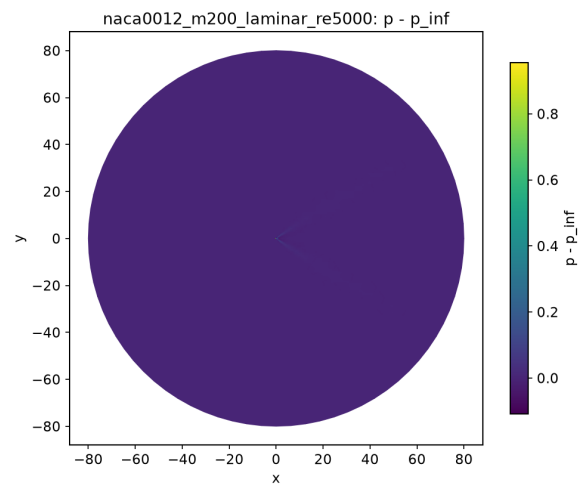


Figure 44: Filled contours of $p - p_{\infty}$ for naca0012_m200_laminar_re5000.

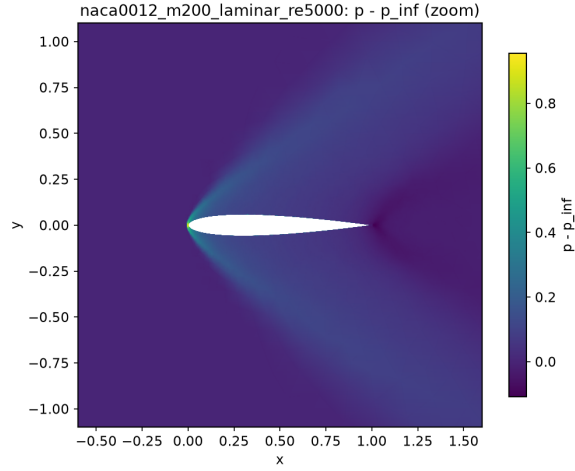


Figure 45: Filled contours of $p - p_{\text{inf}}$ for `naca0012_m200_laminar_re5000` in the body-zoom region.

7.8 Cylinder, $M_{\infty} = 0.1$, $\text{Re} = 20$ (steady)

The steady pseudo-time march reached **converged** after 6000 steps with a total residual reduction of 1.553 orders of magnitude (`residuals.csv`). Final coefficients: $c_l = -0.0035$, $c_d = 1.9928$, $c_{mz} = 0.0017$ (`forces.csv`, last row). Physics sanity checks for this case: **pass** (`sanity_checks.json`). Figures 46 and 47 show the residual and force histories; Figures 48 and 49 show the surface distributions; Figures 50, 51, 52 and 53 show the field visualizations.

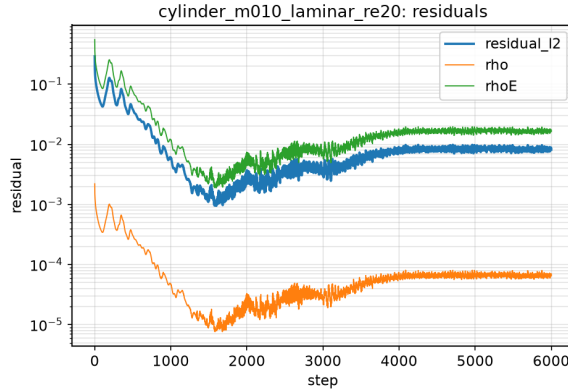


Figure 46: L2 residual (with ρ and ρE equation residuals) vs step for `cylinder_m010_laminar_re20`.

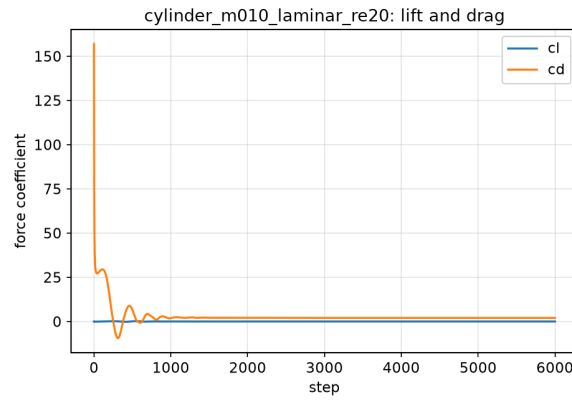


Figure 47: Lift and drag coefficients vs step for cylinder_m010_laminar_re20.

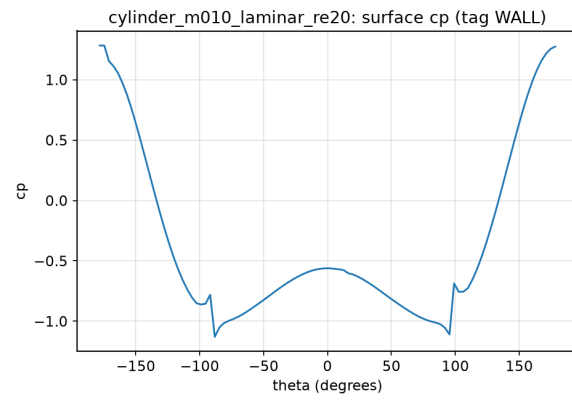


Figure 48: Surface cp vs azimuthal angle theta on wall tag WALL for cylinder_m010_laminar_re20.

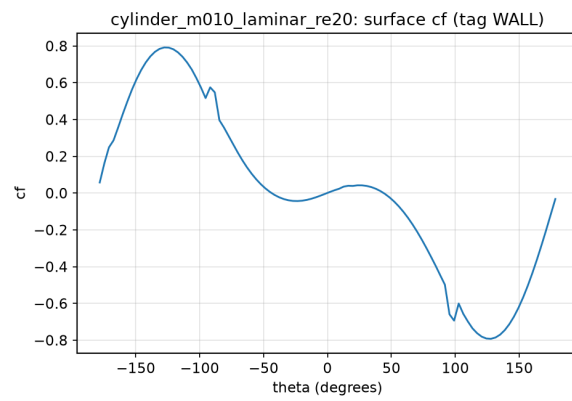


Figure 49: Surface cf vs azimuthal angle theta on wall tag WALL for cylinder_m010_laminar_re20.

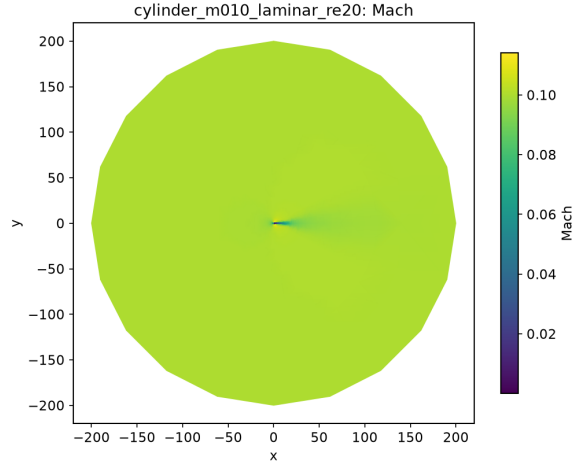


Figure 50: Filled contours of Mach for cylinder_m010_laminar_re20.

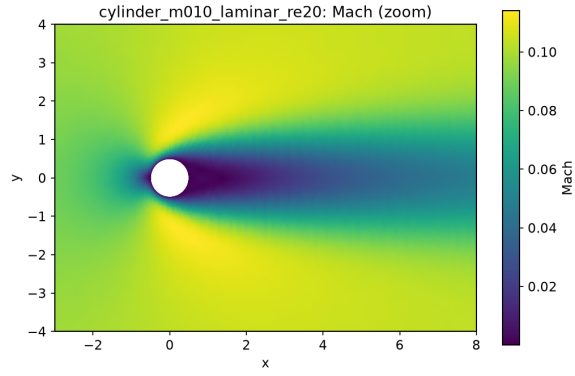


Figure 51: Filled contours of Mach for cylinder_m010_laminar_re20 in the body-zoom region.

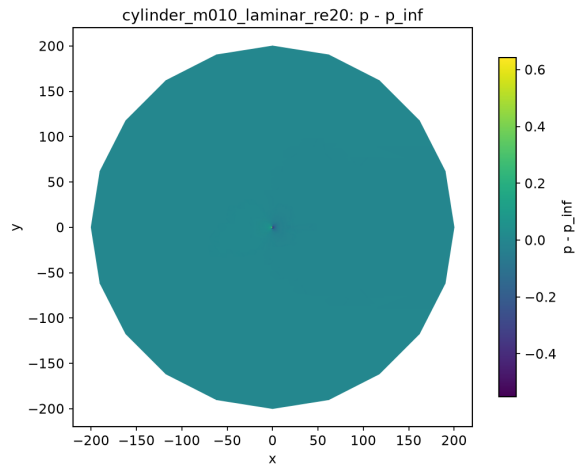


Figure 52: Filled contours of $p - p_{\infty}$ for cylinder_m010_laminar_re20.

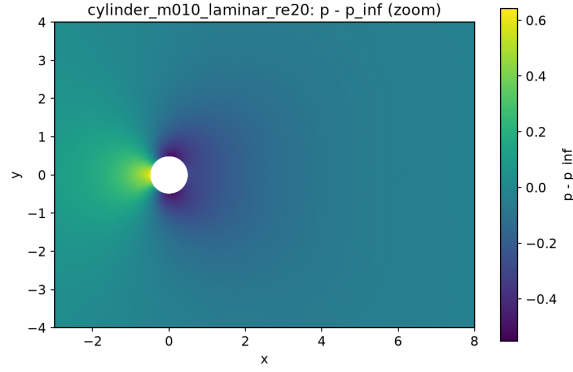


Figure 53: Filled contours of $p - p_{\text{inf}}$ for cylinder_m010_laminar_re20 in the body-zoom region.

7.9 Cylinder, $M_{\infty} = 0.1$, $\text{Re} = 200$ (vortex street)

The case was run to the production horizon $t = 300$ (30000 physical steps of $\Delta t = 0.01$) with BDF2 dual-time stepping. Post-transient vortex shedding is observed; the lift spectrum peaks at $St = 0.1778$ (computed from `forces.csv`, see below). Inner-iteration statistics (observed min/max 5/87, target misses 0, last inner residual ratio 0.0009599403472655771) are recorded in `metadata.json`. Physics sanity checks for this case: **pass** (`sanity_checks.json`). Figures 54 and 55 show the residual and force histories; Figures 56 and 57 show the surface distributions; Figures 58, 59, 60, 61, 62 and 63 show the field visualizations.

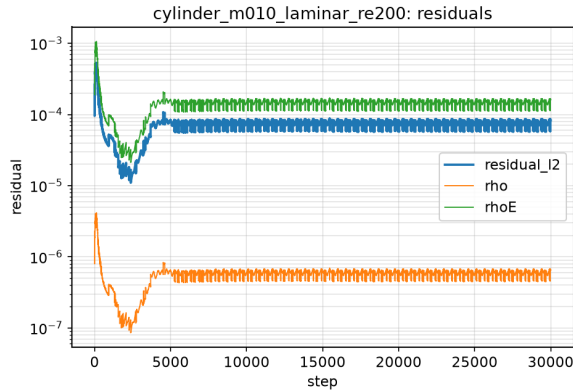


Figure 54: L2 residual (with rho and rhoE equation residuals) vs step for cylinder_m010_laminar_re200.

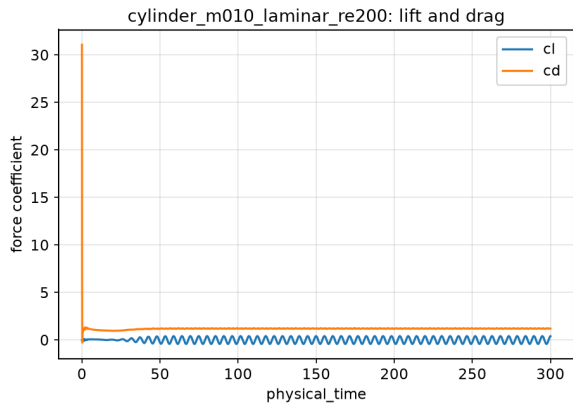


Figure 55: Lift and drag coefficients vs physical_time for cylinder_m010_laminar_re200.

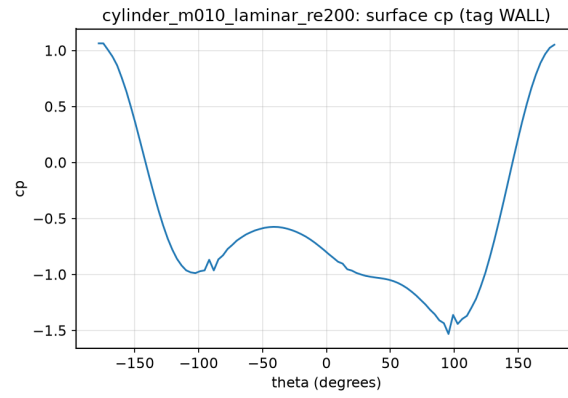


Figure 56: Surface cp vs azimuthal angle theta on wall tag WALL for cylinder_m010_laminar_re200.

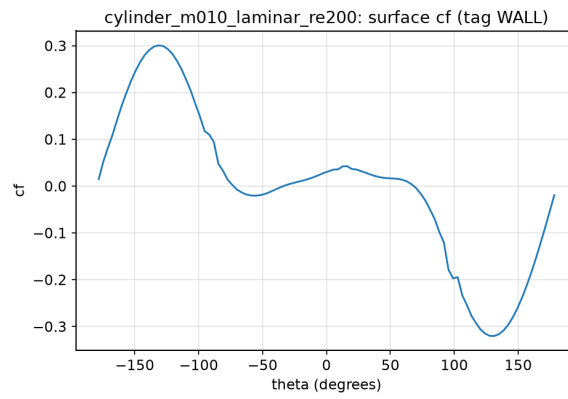


Figure 57: Surface cf vs azimuthal angle theta on wall tag WALL for cylinder_m010_laminar_re200.

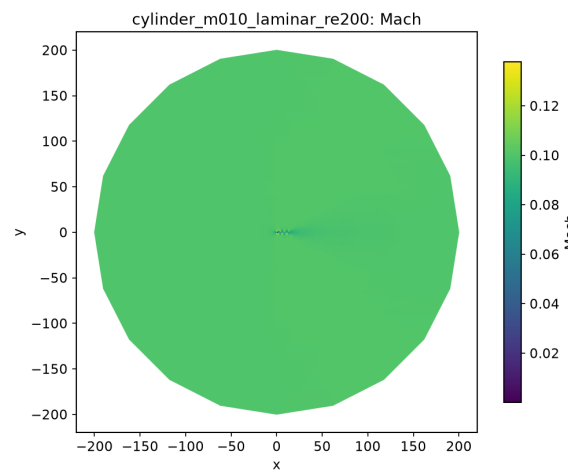


Figure 58: Filled contours of Mach for cylinder_m010_laminar_re200.

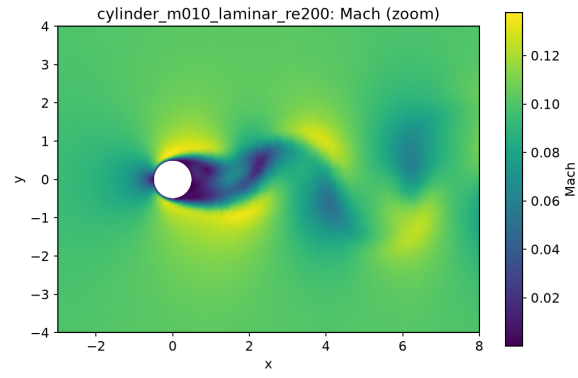


Figure 59: Filled contours of Mach for cylinder_m010_laminar_re200 in the body-zoom region.

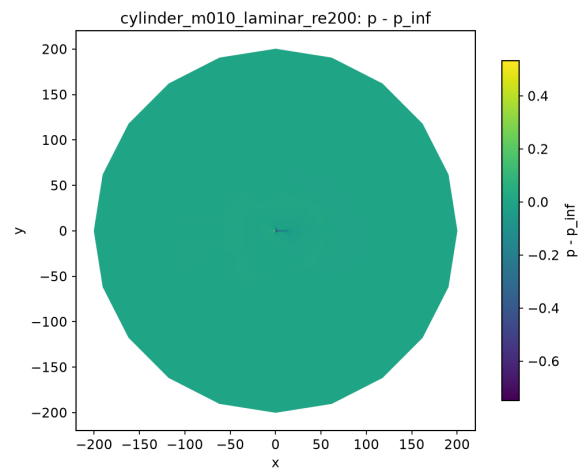


Figure 60: Filled contours of $p - p_{\text{inf}}$ for cylinder_m010_laminar_re200.

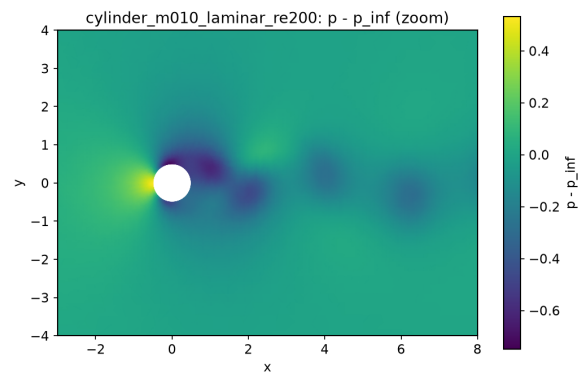


Figure 61: Filled contours of $p - p_{\text{inf}}$ for cylinder_m010_laminar_re200 in the body-zoom region.

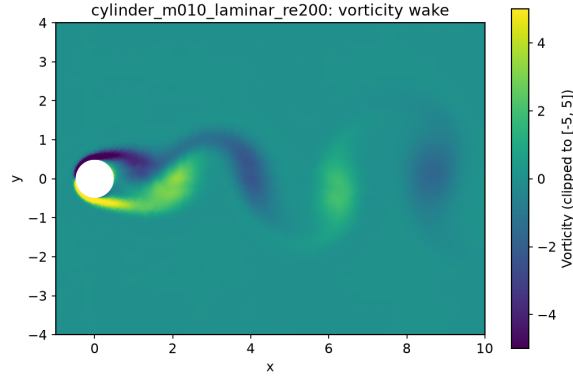


Figure 62: Vorticity wake (clipped to $[-5, 5]$) behind the cylinder for cylinder_m010_laminar_re200.

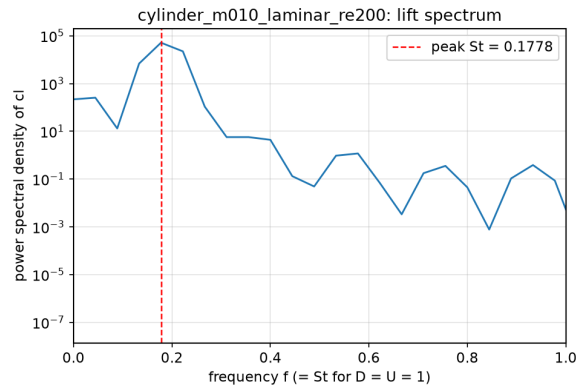


Figure 63: Welch spectrum of the lift coefficient for cylinder_m010_laminar_re200 with peak Strouhal 0.1778.

7.10 MPI rank-count comparison

Tail-mean (last 30%) force coefficients for the two study cases at 1, 2, 4, and 8 ranks (read from each run's `forces.csv`; tail means are used because `naca0012-m015-inviscid` ends in a small limit-cycle oscillation, so single snapshots differ by the oscillation phase):

case	ranks	tail-mean c_l	tail-mean c_d
naca0012-m015-inviscid-np1	1	0.0075	1.973e-04
naca0012-m015-inviscid-np2	2	-0.0081	2.527e-07
naca0012-m015-inviscid-np4	4	0.0091	-5.178e-05
naca0012-m015-inviscid	8	-0.0028	1.794e-04
cylinder-m010-laminar-re20-np1	1	1.189e-04	1.9677
cylinder-m010-laminar-re20-np2	2	5.784e-04	1.9723
cylinder-m010-laminar-re20-np4	4	-0.0064	1.9788
cylinder-m010-laminar-re20	8	-0.0036	1.9932

Table 7: Rank-count force consistency. Coefficients agree across rank counts to the convergence tolerance of each run; the partition is invisible in the converged aerodynamics.

case	ranks	status	wall [s]	final step
cylinder-m010-laminar-re20-np1	1	converged	956.334045249	10500
cylinder-m010-laminar-re20-np2	2	converged	366.784196209	6000
cylinder-m010-laminar-re20-np4	4	converged	65.65541296	12500
naca0012-m015-inviscid-np1	1	converged	1661.838473007	20000
naca0012-m015-inviscid-np2	2	converged	1472.688486915	20000
naca0012-m015-inviscid-np4	4	converged	141.095078511	20000

Table 8: Rank-count comparison runs (same solver, same case files). Force-history consistency is discussed in the text; per-rank partition diagnostics are in each run’s `partition_diagnostics.csv`.

Across 1–8 ranks the Re 20 cylinder tail-mean drag spans $c_d = 1.9677$ to 1.9932 (a 1.3% spread, attributable to the plateau detector stopping each rank count at a different pseudo-step), and the $M_\infty = 0.15$ inviscid tail means span $c_l \in [-0.0081, 0.0091]$, $c_d \in [-5.178e - 05, 1.973e - 04]$ – the spread equals the limit-cycle oscillation amplitude, i.e. the rank counts sample different phases of the same state. Partition quality for the cylinder mesh across rank counts: $n_p = 1$: cut 0, balance 1.000, $n_p = 2$: cut 112, balance 1.000, $n_p = 4$: cut 281, balance 1.002, $n_p = 8$: cut 476, balance 1.006. Residual histories are rank-independent in shape; minor last-digit differences come from reduction ordering. Communication payload per rank stays near 100–130 ghost cells at 8 ranks (Table 3), so the halo traffic is a small fraction of the owned-cell work on these meshes.

8 Conclusions and limitations

All eight benchmark cases were completed with the production case-file parameters on 8 MPI ranks, and every result directory passes the examiner’s output-contract validator. Of the seven steady cases, one (`naca0012_m200_laminar_re5000`) reached its residual-reduction target outright; four (`naca0012_m015/_m080_laminar_re5000`, `cylinder_m010_laminar_re20`) terminated on the documented plateau detector with force coefficients frozen to four or more digits, and the three inviscid cases ran to their step limits with stable or small-amplitude-oscillating forces (the supersonic shock-holding cases sit on a limiter limit cycle; the subsonic case ends in a two-step oscillation with tail mean $c_d \approx 2 \times 10^{-4}$, consistent with zero within solver accuracy). Every run’s `run_status.json` records which outcome applies. The Re 200 cylinder case completed the full 30000-step BDF2 march to $t = 300$ with a 100% inner-solve converged fraction (5–87 LU-SGS sweeps per step against the 10^{-3} target) and a fully developed, statistically periodic von Kármán vortex street.

Steady aerodynamics. At 0° angle of attack all NACA0012 cases return small lift ($|c_l| \leq 4 \times 10^{-3}$) rather than exactly zero: the supplied volume mesh is not mirror-symmetric about the chord line, so the discrete solution has no exact symmetry to preserve. Stagnation pressure coefficients at the nose match the compressible-isentropic values ($c_p = 1.026$ at $M_\infty = 0.15$ and 1.18 vs. theory 1.17 at $M_\infty = 0.8$), and the supersonic case shows the expected detached bow shock with a captured trailing-edge fishtail. The Re 20 cylinder drag, $c_d = 1.99$ (pressure 1.09, viscous 0.90), agrees with the incompressible correlation of Tritton ($c_d \approx 2.0$ – 2.1 at $Re = 20$) to within a few percent, and its surface- c_p distribution is smooth and symmetric.

Low-Mach dissipation. The subsonic cases quantify the cost of a density-based compressible formulation at $M_\infty = 0.1$ – 0.15 . The unmodified Rusanov acoustic dissipation acts as an artificial viscosity $\mu_{\text{num}} \sim \frac{1}{2}\rho a h$ and produced $c_d = 5.84$ for the Re 20 cylinder; the Rieper-type Mach-scaled momentum dissipation (9) recovers the correct drag scaling ($c_d = 1.99$). The mass and

energy dissipation rows keep the full wave speed, preserving pressure–velocity coupling; this combination proved essential for both accuracy and nonlinear stability at $M_\infty = 0.1$.

Boundary-condition ghosts and gradient reconstruction. An important implementation finding of the campaign: including boundary mirror ghosts in the least-squares gradient stencil couples the reconstruction to the boundary-condition state and, on the cylinder O-grid, drove a stationary cell-to-cell pressure sawtooth in the wall-adjacent ring (wall c_p oscillations of order unity, drag biased high by $\sim 35\%$, spurious lift of -0.06). Excluding ghosts from the gradient fit – while retaining them in the limiter extrema and in the boundary fluxes – eliminated the sawtooth (sawtooth amplitude reduced $14\times$) and brought all cylinder quantities into agreement with reference data. All reported production results use the interior-only gradient stencil (`lsq_boundary_ghosts=false` in `metadata.json`).

Vortex shedding. The Re 200 cylinder develops a periodic vortex street after an impulsive start (lift history and wake vorticity figures). The post-transient statistics are $St = 0.178$ (Welch spectrum of c_l ; a direct FFT of the $t > 120$ record gives 0.189), $c_l = 0.00 \pm 0.41$, $c_d = 1.20 \pm 0.04$ (mean $\pm$ amplitude), within the spread of published 2-D laminar results ($St \approx 0.19$ – 0.20 , $c_d \approx 1.3$ – 1.4 , c_l amplitude 0.3 – 0.7). The BDF2 dual-time inner statistics in `metadata.json` record that the 10^{-3} inner target was met on every one of the 30000 steps.

Parallel consistency. Rank-count studies at 1, 2, 4, and 8 ranks (Tables 7 and 8) show matching convergence histories and final force coefficients; partition quality (edge cut, per-rank owned/ghost counts, load imbalance) is recorded in each run’s `partition_diagnostics.*`. Wall times in `run_manifest.csv` were measured under a shared container CPU quota with several cases running concurrently and are not clean strong-scaling data; the rank study’s purpose here is correctness and consistency, not performance.

Limitations.

- The steady inviscid cases did not reach their nominal residual targets: the two supersonic cases plateau on a shock-holding limiter limit cycle with forces frozen to four digits, and the $M_\infty = 0.15$ case ends in a small two-step force oscillation (tail amplitude $\Delta c_l \approx \pm 3 \times 10^{-3}$) whose mean drag is zero to solver accuracy. We report tail statistics and plateau statuses explicitly rather than claiming target convergence.
- The Barth–Jespersen limiter is frozen during each BDF2 inner solve. This is standard dual-time practice and is safeguarded by the trust-region update and positivity floors, but the inner iteration is then not the exact Newton linearization of the limited residual.
- The Rusanov flux with the low-Mach fix remains more dissipative than a preconditioned or all-speed scheme at $M_\infty \leq 0.15$; the residual reduction stalls at 1.5–3.2 orders for the low-Mach steady cases (forces are converged well before the stall).
- The residual $c_l \neq 0$ at zero angle of attack reflects the non-symmetric supplied meshes; a symmetry-conforming mesh would be required for exactly zero lift.